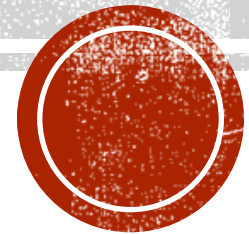


IMMIGRATION MANAGEMENT SYSTEM RESTFUL API

Technologies: Java, Spring Boot, MySQL, Lombok , JPA, Postman

Done by : Shahd AL Abdali



PROJECT OVERVIEW

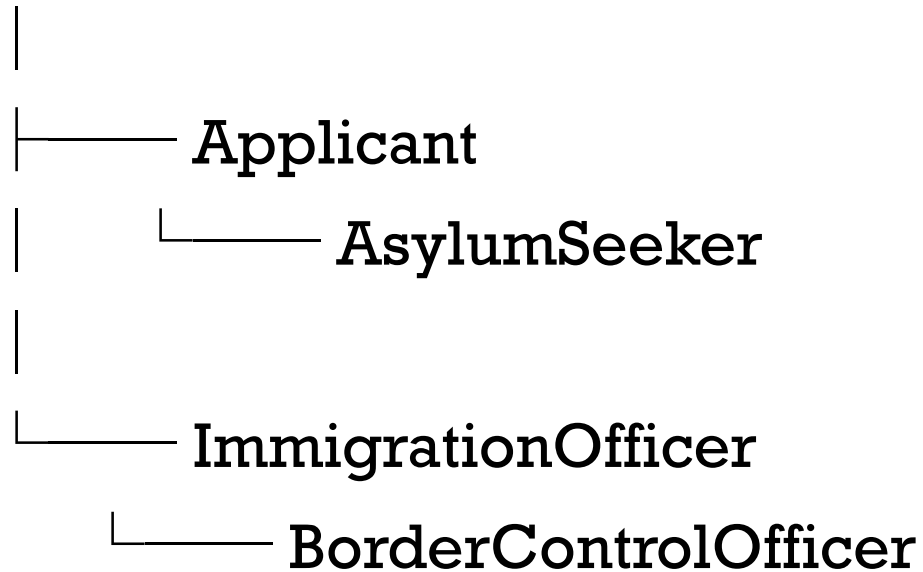
- **Project Description**
- The Immigration System is designed to manage:
 - Immigration Centers
 - Applicants
 - Immigration Officers
 - Visa Applications
 - Interviews
- The system automates visa processing and interview scheduling while enforcing business rule



OOP DESIGN

■ Inheritance Structure

Person

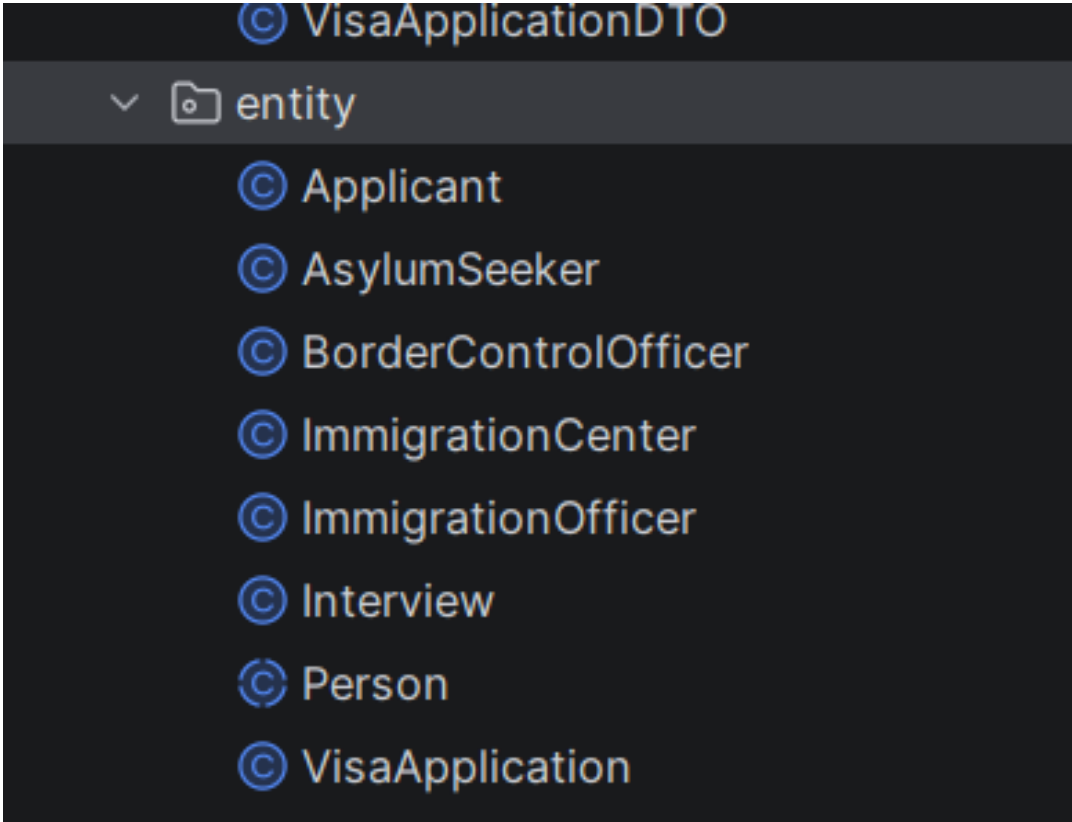


OOP Concepts Used

- Inheritance
- Method Overloading
- DTO Pattern
- Exception Handling



DATABASE ENTITIES



© VisaApplicationDTO

▼ entity

- © Applicant
- © AsylumSeeker
- © BorderControlOfficer
- © ImmigrationCenter
- © ImmigrationOfficer
- © Interview
- © Person
- © VisaApplication

- **Main Entities**
- Person
- Applicant
- AsylumSeeker
- ImmigrationOfficer
- BorderControlOfficer
- ImmigrationCenter
- VisaApplication
- Interview



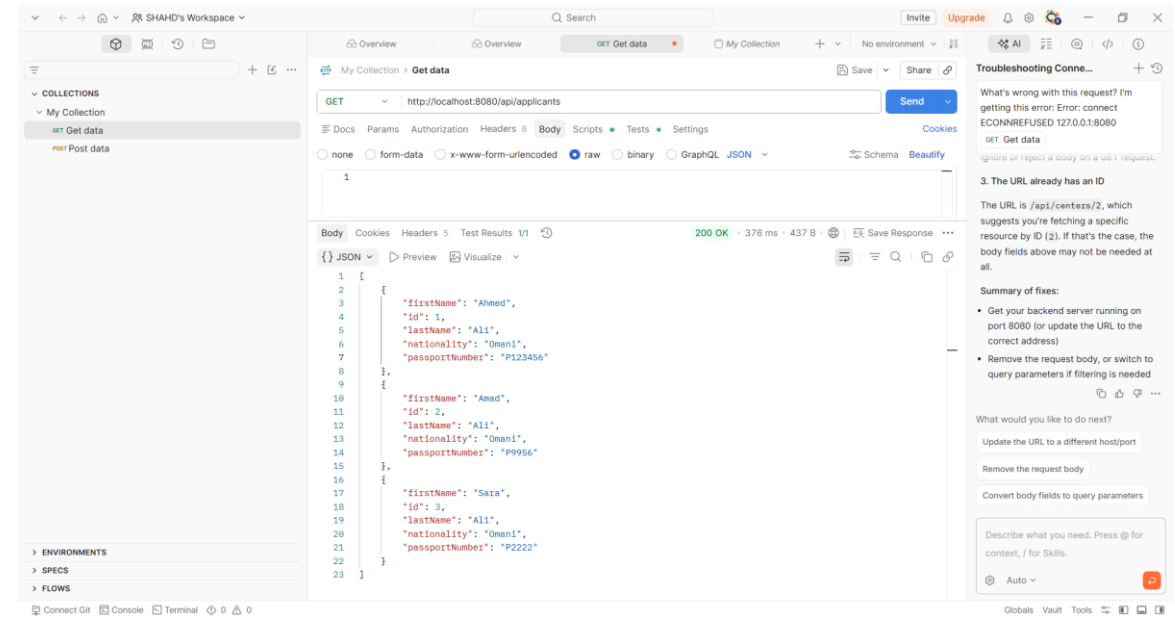
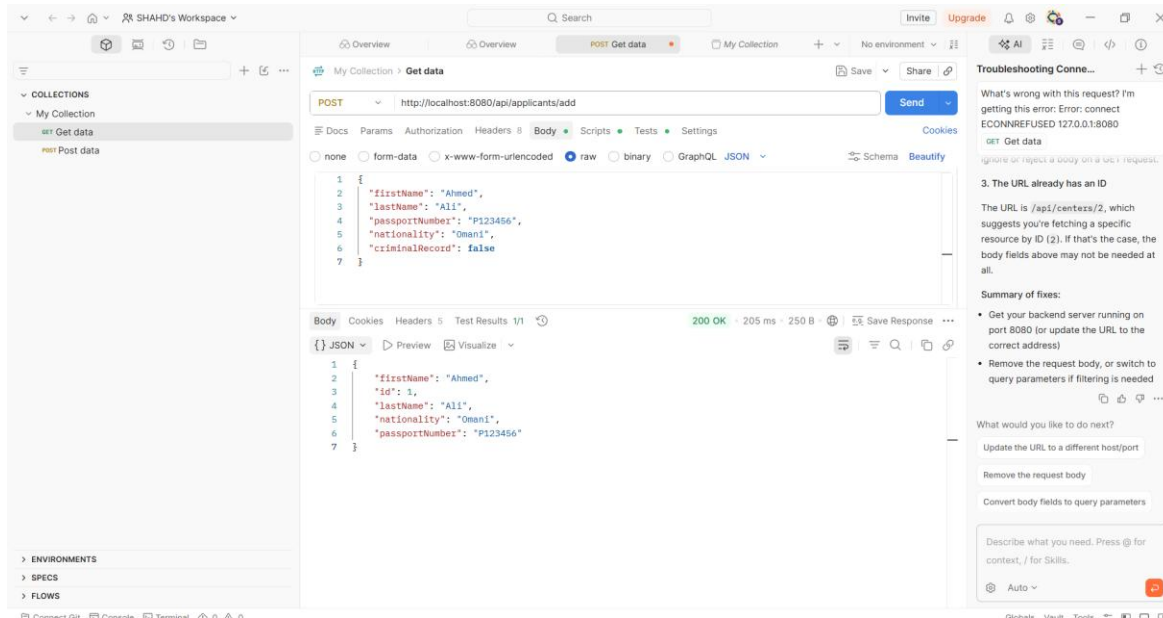
PROJECT STRUCTURE

- entity
- repository
- service
- controller
- dto
- Exception



APPLICANT ENDPOINTS

- Create Applicant Endpoint
- DTO hides sensitive/internal fields.
- Criminal record flag is stored and validated.



OFFICER ENDPOINTS

My Collection > Get data

GET http://localhost:8080/api/officers/1

Body

```
1 {
2   "badgeNumber": "OF001",
3   "firstName": "Shahd",
4   "id": 1,
5   "lastName": "Ahmed",
6   "officerRank": "Junior Officer"
7 }
```

200 OK · 18 ms · 257 B

Body Cookies Headers 5 Test Results 1/1

JSON Preview Visualize

Convert Get data into parameterized tests
Create a request-level test for POST
Set up an environment and baseUri variables

Describe what you need. Press @ for context, / for Skills.

Auto

My Collection > Get data

PUT http://localhost:8080/api/officers/1/promote?rank=Senior Officer&clearance=4

Body

```
1 {
2   "badgeNumber": "OF001",
3   "firstName": "Shahd",
4   "id": 1,
5   "lastName": "Ahmed",
6   "officerRank": "Senior Officer"
7 }
```

200 OK · 8 ms · 257 B

Body Cookies Headers 5 Test Results 1/1

JSON Preview Visualize

Adding a visualization s...

Help me add a visualization script

Retrieved details for request Get data from tab

I can see your Get data request. It's a PUT request to promote an officer, and the last response returned this JSON:

```
{
  "badgeNumber": "OF001",
  "firstName": "Shahd",
  "id": 1,
  "lastName": " "
}
```

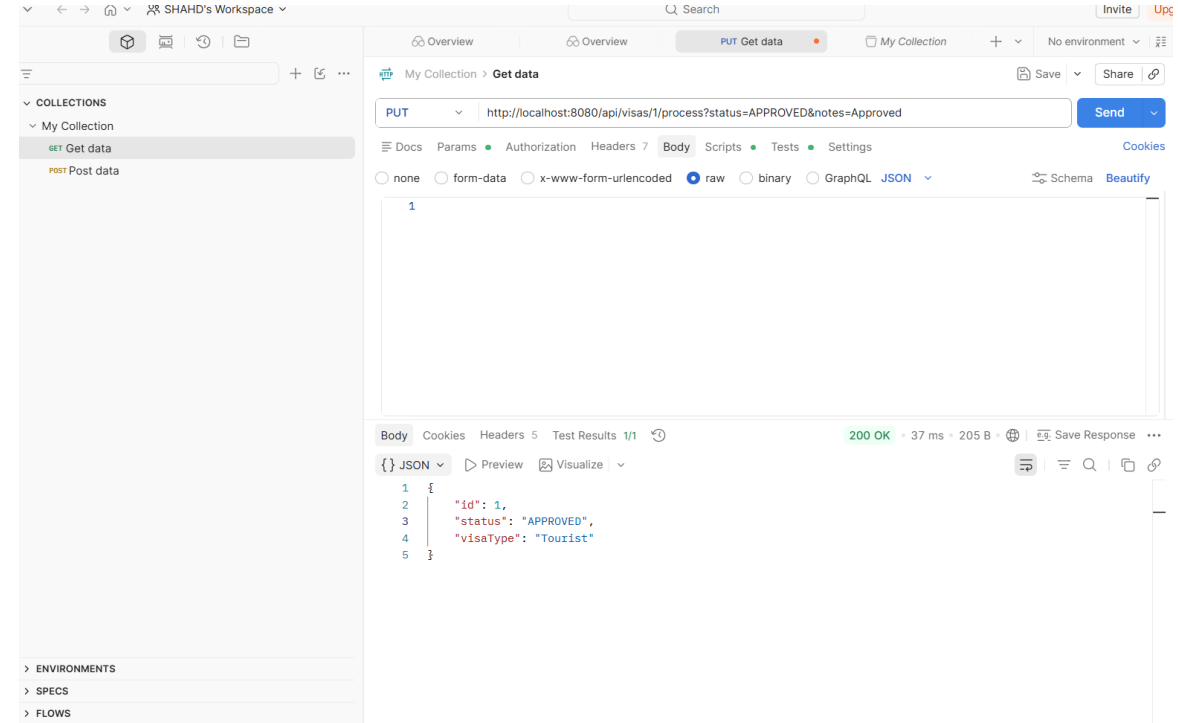
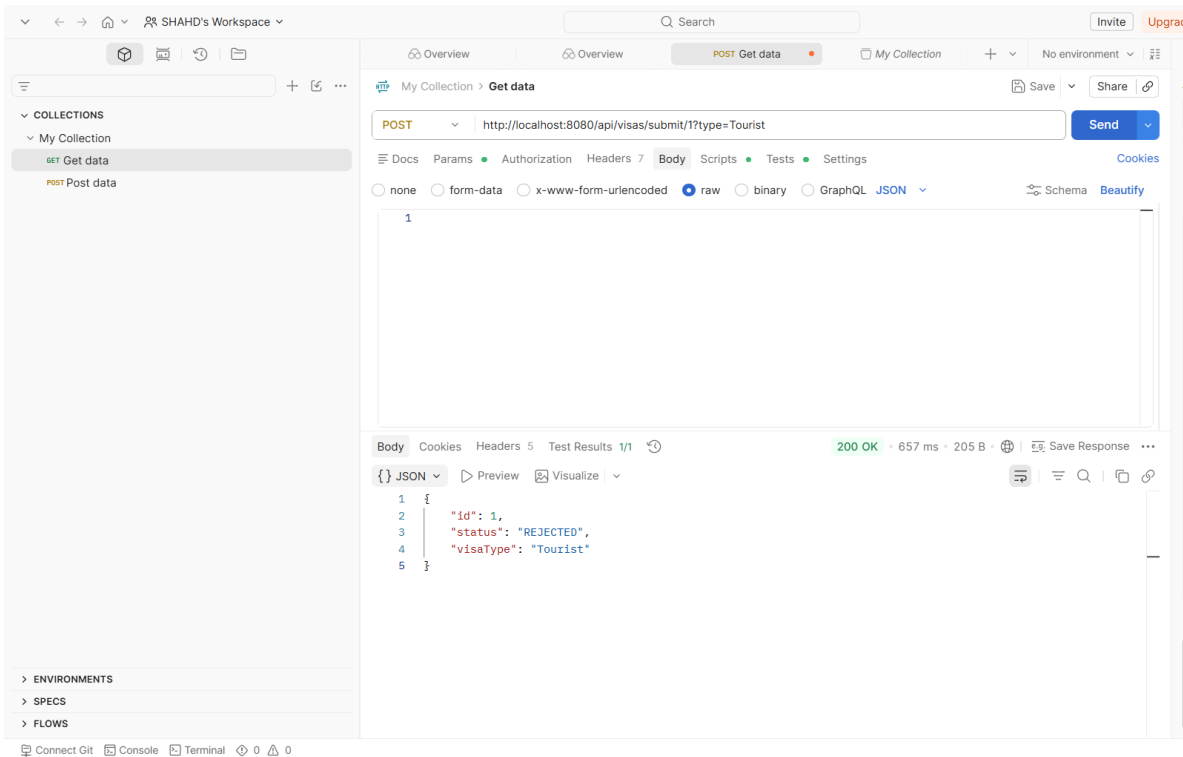
Describe what you need. Press @ for context, / for Skills.

Auto



VISA APPLICATION ENDPOINTS

- Submit Visa
- Process Visa (Approve / Reject)



GET VISAS BY STATUS

The screenshot displays the Postman interface for a REST client. On the left, the 'COLLECTIONS' sidebar shows 'My Collection' with two items: 'GET Get data' (selected) and 'POST Post data'. The main workspace shows the 'GET Get data' request configuration. The URL is 'http://localhost:8080/api/visas/status/APPROVED'. The 'Body' tab is active, showing a single JSON object. The response status is '200 OK' with a response time of 823 ms and a size of 207 B. The response body is a JSON array containing one object with the following fields: 'id' (1), 'status' (APPROVED), and 'visaType' (Tourist).

Overview Overview GET Get data My Collection + No environment

My Collection > GET data Save Share

GET http://localhost:8080/api/visas/status/APPROVED Send

Docs Params Authorization Headers 6 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

1

Body Cookies Headers 5 Test Results 1/1 200 OK • 823 ms • 207 B Save Response

```
{
  "id": 1,
  "status": "APPROVED",
  "visaType": "Tourist"
}
```



SHAHD's Workspace

Overview Overview POST Get data My Collection No environment

My Collection > Get data

POST http://localhost:8080/api/interviews/schedule/1/1?date=2025-06-20&purpose=Visa Interview Send

Docs Params Authorization Headers 7 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

1

Body Cookies Headers 4 Test Results 0/1 400 Bad Request 1.78 s 300 B Save Response

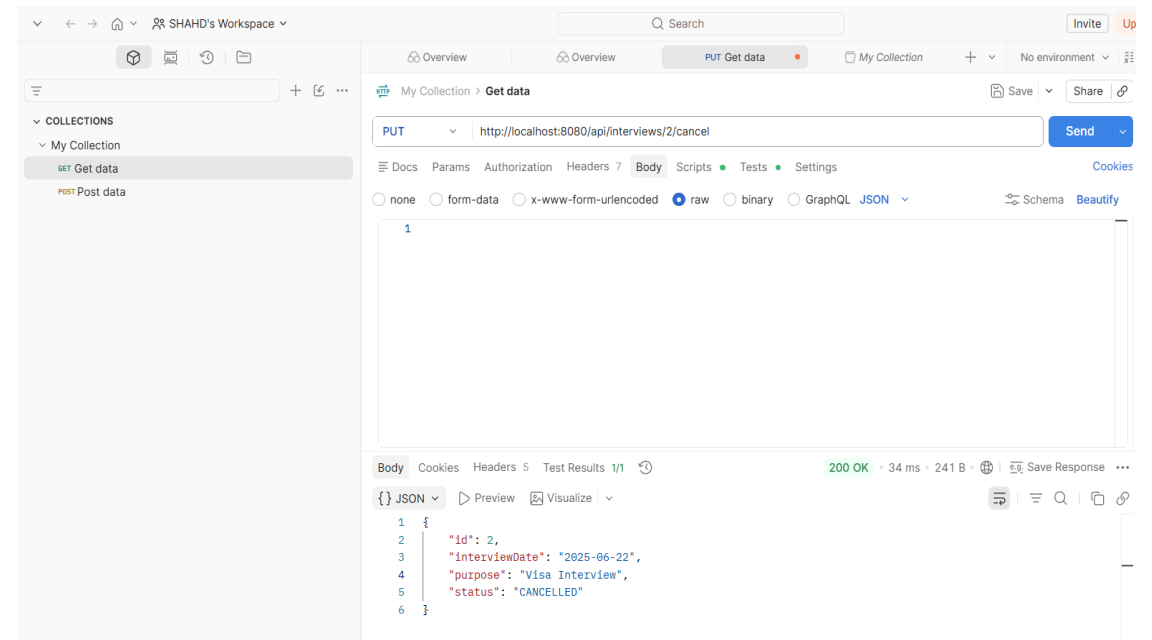
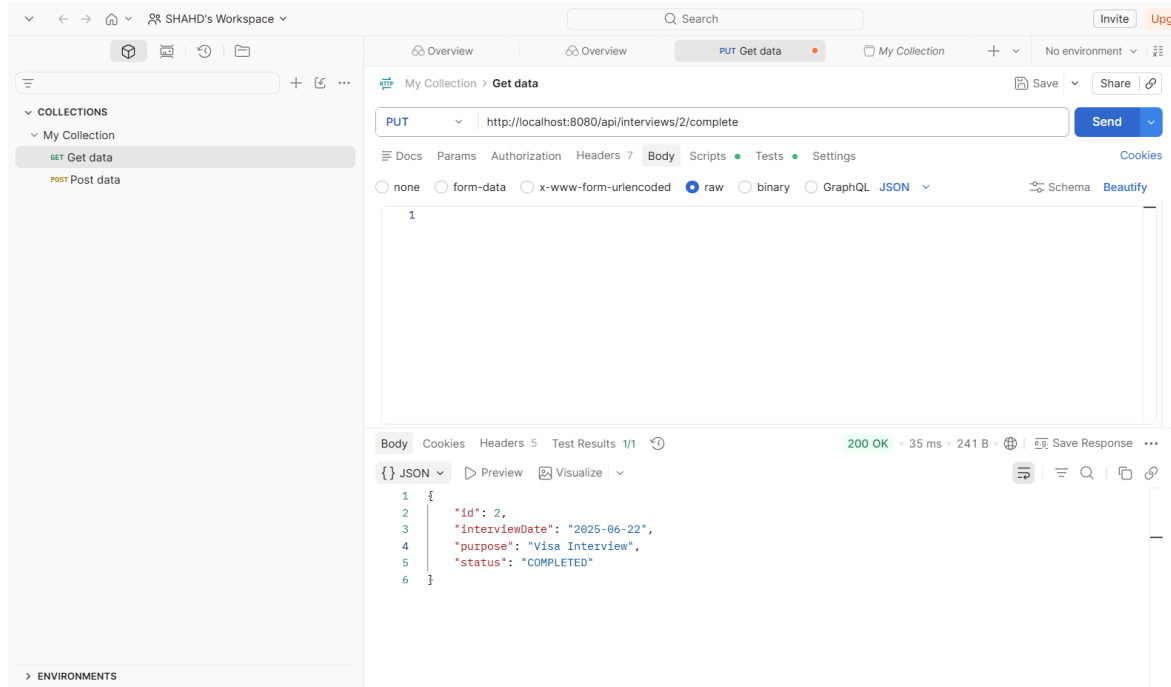
{ } JSON Preview Debug with AI

```
1 {
2   "error": "Bad Request",
3   "message": "Officer is double-booked!",
4   "path": null,
5   "status": "400 BAD_REQUEST",
6   "statusCode": 400,
7   "timestamp": "2026-06-16T07:32:46.339Z"
8 }
```

DOUBLE BOOKING TEST



COMPLETE INTERVIEW - CANCEL



GET OFFICER SCHEDULE

The screenshot displays the Postman interface for a REST client. On the left, the 'COLLECTIONS' sidebar shows 'My Collection' expanded, with 'GET Get data' selected. The main workspace shows the 'GET Get data' request configured with the URL 'http://localhost:8080/api/interviews/officer/1/date/2025-06-22'. The 'Body' tab is active, showing a 'raw' response. The status bar at the bottom indicates a '200 OK' response with a 22 ms duration and 243 B of data. The response body is a JSON array containing one object with the following details:

```
1 [
2   {
3     "id": 2,
4     "interviewDate": "2025-06-22",
5     "purpose": "Visa Interview",
6     "status": "CANCELLED"
7   }
8 ]
```



DATABASE SEED ENDPOINT

The screenshot displays the Postman API client interface. On the left, the 'COLLECTIONS' sidebar shows 'My Collection' with two items: 'GET Get data' and 'POST Post data'. The main workspace is titled 'My Collection > Get data' and shows a 'POST' request to 'http://localhost:8080/api/seed'. The request body is set to 'raw' and contains the number '1'. The status bar at the bottom indicates a successful response: '200 OK • 401 ms • 193 B'. Below the status bar, the response body is shown as '1 Database seeded successfully!'. On the right, the 'Adding a visualization s...' sidebar is visible, showing a JSON response for the 'Get data' endpoint:

```
{  "badgeNumber": "OF",  "firstName": "Shah",  "id": 1,  "lastName": "Ahmed",  "officerRank": "Se Officer"}
```



MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

apis8db

banking

immigration

Tables

applicant

applicant_interviews

applicant_visa_applications

Columns

Indexes

Foreign Keys

Triggers

immigration_center

immigration_center_officers

Columns

Indexes

Foreign Keys

Triggers

immigration_officer

immigration_officer_interviews

interview

visa_application

Views

Stored Procedures

Functions

sys

Administration Schemas

applicant 2 immigration_officer 3 visa_application 4 interview 5 immigration_center

Apply Revert Context Help Snippets

Information

Table: applicant

Columns:

dtype

id

email

first_name

gender

last_name

varchar(31)

bigint AI PK

varchar(255)

varchar(255)

varchar(255)

varchar(255)

varchar(255)

Object Info Session

1 SELECT * FROM applicant;

2 SELECT * FROM immigration_officer;

3 SELECT * FROM visa_application;

4 SELECT * FROM interview;

5 SELECT * FROM immigration_center;

Result Grid

	id	daily_capacity	location_country	name	type
1	200		Iraq	Baghdad Center	Embassy
2	200		Oman	Muscat Center	Embassy
3	200		Oman	Muscat Center	Embassy
4	100		Oman	Salalah Center	Border
5	200		Oman	Muscat Center	Embassy
6	100		Oman	Salalah Center	Border

Automatic context help is disabled. Use the toolbar for the current caret position or to toggle all.

Output

Action Output

#	Time	Action	Message
33	13:06:22	SELECT * FROM immigration.applicant LIMIT 0, 1000	11 row(s) returned
34	13:08:30	SELECT * FROM applicant LIMIT 0, 1000	11 row(s) returned
35	13:08:30	SELECT * FROM immigration_officer LIMIT 0, 1000	7 row(s) returned
36	13:08:30	SELECT * FROM visa_application LIMIT 0, 1000	1 row(s) returned
37	13:08:30	SELECT * FROM interview LIMIT 0, 1000	2 row(s) returned
38	13:08:30	SELECT * FROM immigration_center LIMIT 0, 1000	6 row(s) returned

DATABASE RESULTS

